William Henry Melvin

540-688-8237 | wmhenrymelvin@gmail.com | linkedin.com/in/william-melvin-b56901348 | wmhenrymelvin.github.io

EDUCATION

James Madison University **Harrisonburg, VA**

Bachelor of Science, Integrated Science & Technology | Concentration: Manufacturing | Minor: Global Supply Chain Management
Junior | Expected Graduation: May 2028

Relevant Coursework: Thermodynamics, Applied Physics, Applied Statistics, Sustainable Manufacturing Systems, Programming, Biotechnology, Energy Issues, Calculus I & II

ENGINEERING EXPERIENCE

Engineering Intern, Ryzing Technologies | *Staunton, VA* **May 2026 – August 2026**

- Performed an ROI analysis on a fabric-cutting machine for SWIFTS Casualty Collection Point (CCP) and Casualty Protection Unit (CPU) production, comparing hand cutting (859 cuts per set, about 41.7 minutes) with the machine's rated 140 pieces per minute across a 100-set volume, an estimated 85% cut in cutting time. Projected return fell below the required threshold, so the purchase was not pursued.
- Created multi-view engineering drawings (flat, rolled, isometric) with bills of materials for CCP/CPU circumferential straps in Onshape. Redesigned the strap pattern for repeatable, sewable manufacturing and sewed a run of finished straps by hand.
- Assisted in fabricating air beams for CCP/CPU units (measuring, adhesive bonding), documented the process step by step so others could repeat it, and helped build 5 complete sets.
- Performed scale-model testing for a military vehicle stability project (ISAS): helped build a torsion pendulum rig to measure moment of inertia and radius of gyration at crew-loaded and gross vehicle weight, checked results against reference figures, and compiled the findings into a presentation for Navy personnel.
- Machined parts for NASA-related projects.
- Sat in on three engineering program reviews with military and NASA personnel.

ENGINEERING PROJECTS

Electronic Throttle Control System with Closed-Loop Speed Feedback | *team of two*

- Co-designed an Arduino-based electronic throttle controller for engine wind testing, integrating Hall effect sensors and tachometers for closed-loop speed feedback.
- Machined the engine mount (CNC, mill, lathe, band saw) and reworked the design after the first version failed under real operating conditions.
- Redesigned the actuation from a rigid coupler to a pulley-and-cable system, added a gearbox for torque, and fitted limit switches to protect the throttle body.
- Developed and tuned an embedded motion controller (TFT display, SD-card CSV motion profiles) driving a servo-actuated mount. Operation was fully repeatable in bench testing, and final testing awaits a replacement power supply.

TECHNICAL SKILLS

CAD and Drawing: Onshape, Engineering Drawings, Bills of Materials
Manufacturing: Manual CNC, Manual Lathe, Industrial Sewing, 3D Printing
Controls and Electronics: Arduino, Servo Integration, Sensor Integration
Software: Microsoft Excel, Word, PowerPoint

CERTIFICATIONS

Google Project Management; Microsoft 365 Fundamentals, Excel, Word, PowerPoint; CPR/AED and First Aid; DoD Mandatory Controlled Unclassified Information (CUI) Training; Onshape: Sketching, Part Studios, Assemblies, 2D Drawings, Direct Editing

ADDITIONAL EXPERIENCE

Lifeguard, West Side Swim Club **Summers 2023 – 2025**

- Served as full-time lifeguard for three summers, including acting assistant manager.
- Assisted patrons proactively and responded to emergencies; maintained CPR/AED certification.